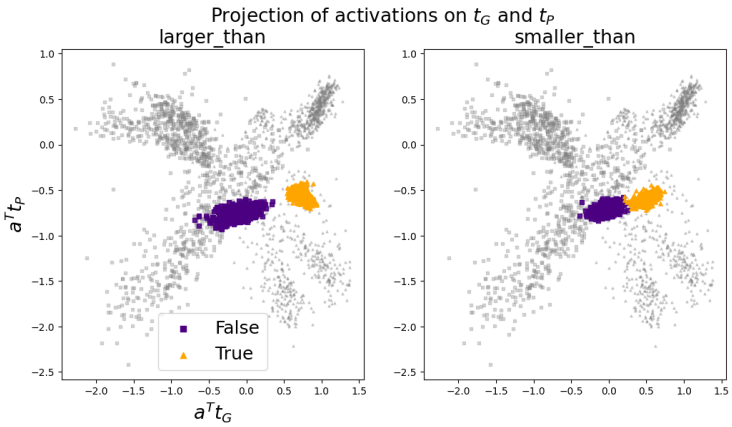

Figure 7: The activation vectors of the `larger_than` and `smaller_than` datasets projected onto $\mathbf{t}_G$ and $\mathbf{t}_P$. In grey: the activation vectors of statements from all affirmative and negated topic-specific datasets.

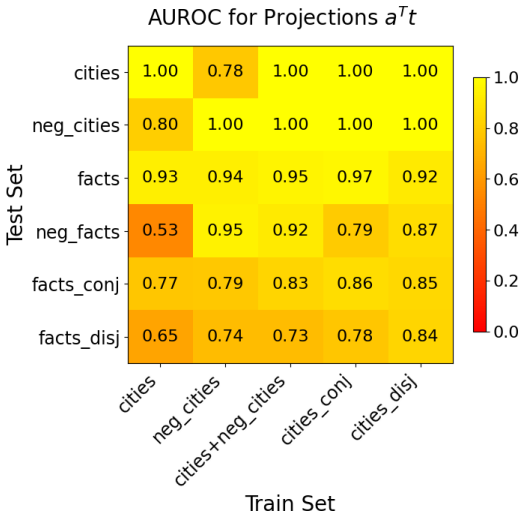

Figure 8: Cross-dataset generalization matrix

We train all truth directions on 80% of the data, evaluating on the held-out 20% if the test and train sets are the same, or on the full test set otherwise.

D Failures modes of the TTPD classifier

In this section, we analyse the failure modes of the TTPD classifier for several datasets. We observed two main failure modes for misclassified statements: In the first failure mode, almost all misclassified statements in a given dataset had the same truth label, while the learned truth direction is still able to separate true from false statements. The reason for these errors is that the bias, learned from other datasets, did not generalize well enough. For example, all $\sim$200 misclassified statements from *cities* had the truth label "False", even though true and false statements separate perfectly along the truth direction $\mathbf{t}_G$, as evidenced by the AUROC of 1.0 in Figure 3. This failure mode also occurred for *neg_cities* and *neg_sp_en_trans*. Below we list a few example statements along with their truth value:

- The city of Bijie is in Indonesia. (False)
- The city of Kalininskiy is not in Russia. (False)
- The Spanish word 'ola' does not mean 'wave'. (False)